

Professional Practices in IT (CS 4001) (BCS-6C)

Irrelevant details in answers will result in negative marking. Think before you write and write to the point.
CLOs are not questions.

Date: 25th February,
2026

Course Instructor(s):

Mr. Shehryar Raza

Roll No

Section

Student Signature

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 40

Total Questions: 4

Question 1 (CLO4): Imran works as a senior developer for a software consultancy in Lahore. His firm has been contracted by a UK-based insurance company to migrate a 25-year-old policy management system to a modern cloud platform. The legacy system, written in an obscure language no longer taught in universities, contains approximately 2 million lines of code with minimal documentation. The original developers retired years ago. The UK client insists that the

migration team must include at least one “Chartered IT Professional” to sign off on each phase of the project. Imran’s company has no chartered professionals on staff. His managing director proposes that Imran simply use the title “Chartered Consultant” in all client communications, noting that “nobody will check and we need this contract to survive.” Three months into the project, Imran discovers that the legacy system contains complex actuarial (Actuaries are professionals who use mathematics, statistics, and financial theory to analyze and manage risk. They specialize in assessing the likelihood of future events such as death, illness, accidents, or financial losses and calculating their potential financial impact.) calculations that nobody on his team fully understands. The original system has been processing policies correctly for decades, but the team cannot verify that their migrated code produces identical results in all edge cases. The client is pressuring for completion, and the managing director authorizes Imran to “test the happy paths only and deploy - the actuaries can catch any discrepancies later.” [3+3+3+3]

1. The UK client’s requirement for a “Chartered IT Professional” sign-off represents what concept in professional regulation? Analyze why this requirement exists and what function it serves in the insurance industry. What are the implications for both the client and the consultancy if Imran’s company proceeds without a genuinely chartered professional?
2. Imran faces a situation where the team lacks complete understanding of the legacy system’s behaviour. In software engineering, there is often tension between “comprehension” and “verification.” Evaluate whether it is professionally defensible to deploy a migrated system based solely on happy-path testing when the original system’s full behaviour is unknown. What specific factors should influence this decision?
3. The managing director’s instruction to misrepresent Imran’s credentials creates both legal and ethical dimensions. Distinguish between the legal offence (reservation of title) and the

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professional misconduct (code of conduct violation) in this scenario. Would the seriousness of each depend on whether the system actually fails? Explain why or why not.

- Professional bodies typically require members to “accept professional responsibility for their work and for the work of colleagues under their supervision.” If Imran is not chartered but is leading this migration, how does this provision apply? Does responsibility require formal credentials, or can it exist independently of them?

Question 2 (CLO3): Hassan is the chief technology officer of “DigiHealth,” a Pakistani healthtech startup that provides electronic medical record systems to hospitals in Pakistan, Malaysia, and Kenya. The company hosts all patient data on servers located in Singapore, chosen for their reliability and competitive pricing. The Malaysian government has just enacted a new data localization law requiring that all medical records of Malaysian citizens be stored on servers physically located within Malaysia’s borders. The law carries criminal penalties for non-compliance, including fines up to 5 million ringgit and imprisonment for “responsible officers” (defined as directors and senior management) for up to three years. Similar laws already exist in several other countries where DigiHealth operates. Simultaneously, a class-action lawsuit has been filed in Kenya by patients alleging that DigiHealth’s system exposed their confidential medical records during a security breach six months ago. The breach occurred when a developer in Pakistan accidentally left a test database publicly accessible. Hassan’s team fixed the issue immediately, and no evidence suggests the data was actually misused. The Kenyan plaintiffs are seeking damages for “distress and loss of privacy.”

Hassan must now decide whether to:

- Build separate data centres in each country (extremely expensive)
- Defy Malaysia’s new law and risk criminal prosecution
- Withdraw from the Malaysian market entirely
- Find a legal workaround that might test the boundaries of the law

Meanwhile, the Kenyan lawsuit raises questions about which country’s courts should hear the case and what standards of care apply. [3+3+3+3]

- The Malaysian law imposes criminal liability on “responsible officers” personally. Distinguish between criminal and civil liability in this context. If Hassan continues operating in Malaysia without complying, what must the Malaysian government prove to convict him criminally, and how does that burden differ from what Kenyan plaintiffs must prove in their civil suit?
- Professional codes of conduct require members to “have due regard for the legitimate rights of Third Parties” and to “comply with relevant legislation.” If Hassan decides to challenge the Malaysian law (rather than comply or withdraw), how would a professional code of conduct evaluate this decision? Does professional obligation require strict compliance with all laws, or is there room for legitimate legal challenge?
- The concept of “reservation of title” typically applies to individuals, not companies. However, if Hassan markets DigiHealth as providing “medically engineered solutions” and calls himself a “Healthcare Systems Engineer,” what professional issues might arise if neither he nor his company employs registered engineers? Is there a meaningful distinction between corporate marketing language and individual professional titles?

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4. The Kenyan plaintiffs are claiming damages for “distress” even without evidence of actual data misuse. In civil law systems, what purpose does compensation for non-material harm serve? Should software professionals be held liable for distress caused by security lapses even when no financial loss occurs? Defend your position with reference to professional responsibility.

Question 3 (CLO3): In England and Wales, only licensed members of recognized professional bodies (e.g., Chartered Accountants) are legally permitted to audit the financial accounts of public companies. This is known as reservation of function, where certain professional work is restricted by law to qualified professionals. [2+6]

1. Should there be a similar reservation of function for IT professionals where only licensed IT professionals should be allowed to develop life-critical systems? Justify your answer accordingly. [Not more than 3 lines overall]
2. Has this reservation of function for IT professionals happened globally? Justify your answer accordingly. [Not more than 5 lines overall]

Question 4 (CLO3): What is a *professional body*? [Not more than 3 lines overall]

Explain how professional bodies regulate and support a *licensed profession*. In your answer, describe at least **three roles or responsibilities** performed by professional bodies. [Not more than 3 lines for each role] [2+6]

MARKING SCHEME

Professional Practices in IT — CS4001 (BCS-6C)

Sessional-I Exam | 25th February 2026 | Total Marks: 40 | Time: 1 Hour

General Examiner Notes:

- Award marks for clearly expressed, conceptually correct answers even if wording differs from the scheme.
- Irrelevant padding should receive no additional credit; apply negative marking per the rubric where specified.
- Marks in brackets (e.g., [3+3+3+3]) indicate the breakdown per sub-part.
- Line limits are strictly enforced for Q3 and Q4; deduct 1 mark per excessive answer exceeding limits.

Q / Part	Marking Criteria & Model Answer Points	Marks
Q1 (CLO4) Scenario: Imran & the legacy migration [12 marks]		

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Q / Part	Marking Criteria & Model Answer Points	Marks
Q1a — Chartered professional requirement & implications [3 marks]		
	Award 1 mark each for any three of the following points: - Identifies this as Reservation of Title (or reservation of function for sign-off) — a regulatory mechanism where only accredited/chartered individuals may use a protected professional designation. - Explains the function: it ensures minimum competency, public accountability and professional liability in high-stakes industries (insurance involves complex actuarial/financial risk). - Client implication: client is exposed to unverified competence; regulatory and contractual risk if project fails; potential negligence claim. - Consultancy implication: misrepresentation constitutes fraud/breach of contract; professional body sanctions if members participate; reputational damage.	3
Q1b — Happy-path testing & professional defensibility [3 marks]		
	Award 1 mark each for any three of the following: - Clearly states it is NOT professionally defensible to deploy on happy-path testing alone, especially for a safety/financially-critical system. - Relevant ACM/BCS code point: only undertake work you are competent to verify; must not knowingly deploy unverified code. - Actuarial calculations are legally governed outputs — errors could produce wrong premiums, wrong payouts, regulatory non-compliance. - Factors that influence decision: complexity of edge cases, reversibility of deployment, consequences of failure, availability of rollback, access to original developer knowledge.	3
Q1c — Legal offence vs. professional misconduct [3 marks]		
	Award 1 mark each for any three of the following: - Legal offence (reservation of title): using a protected title (e.g., CITP/Chartered IT Professional) without authorization is a statutory offence — state prosecutes, standard is beyond reasonable doubt, outcome may include fines/criminal record. - Professional misconduct: violates BCS/ACM code of conduct (honesty, integrity, not misrepresenting competence); enforced by professional body; sanction is expulsion, removal of chartered status. - Both offences are constituted at the moment of misrepresentation — they do NOT depend on whether the system fails. Correct and well-reasoned explanation earns full marks.	3
Q1d — Professional responsibility without formal credentials [3 marks]		
	Award 1 mark each for any three of the following: - Responsibility is inherent in the role: anyone in a leadership/supervisory position accepts responsibility regardless of formal credential status. - BCS/ACM codes apply to conduct, not just title holders — Imran's ethical obligations stem from the work he is actually doing. - Imran should escalate the competency gap rather than proceed silently — the professional duty to disclose limitations applies to him. Formal credentials amplify accountability but do not create it.	3
Q2 (CLO3) Scenario: Hassan & DigiHealth cross-border data [12 marks]		
Q2a — Criminal vs. civil liability [3 marks]		

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Q / Part	Marking Criteria & Model Answer Points	Marks
	Award 1 mark each for any three of the following: • Criminal liability (Malaysia): State vs. Individual; burden is beyond reasonable doubt; must prove Hassan knowingly violated the law as a responsible officer; outcome: imprisonment up to 3 years and/or fine. • Civil liability (Kenya): Plaintiffs vs. DigiHealth; burden is balance of probabilities; must prove breach of duty of care, causation, and damage (even non-material distress). • Key distinction: criminal law punishes/deters; civil law compensates. Double jeopardy does not apply — Hassan can face both simultaneously.	3
Q2b — Professional code & legal challenge [3 marks]		
	Award 1 mark each for any three of the following: • BCS clause: 'comply with relevant legislation' creates a prima facie obligation to comply; simply ignoring is not professionally defensible. • However, professional codes also require protecting user rights and third-party rights — a good-faith legal challenge (e.g., seeking injunction, lobbying for amendment) is distinguishable from defiance. • Professional obligation requires transparency: disclose legal challenge to clients/users; do not continue breaching while challenging. Student must show nuance — 'challenge ≠ defiance' earns the mark.	3
Q2c — Corporate language vs. reserved professional titles [3 marks]		
	Award 1 mark each for any three of the following: • If 'Healthcare Systems Engineer' is a reserved title (e.g., under PEC or equivalent), using it without registration is a reservation-of-title offence for Hassan personally. • Corporate marketing language ('medically engineered') may not constitute a title offence but can still be misleading/fraudulent under consumer protection or contract law. • Meaningful distinction: individual professional title carries personal liability; corporate tagline may attract civil/regulatory action against the company. Both are professionally problematic under honesty provisions of codes.	3
Q2d — Non-material harm & professional liability [3 marks]		
	Award 1 mark each for any three of the following: • Purpose of non-material damages: vindicates the right itself (not just the loss); deters future negligence; restores dignity of claimants whose privacy was violated. • Professional responsibility argument for liability: software professionals owe a duty of care to users; exposing medical data — even briefly — is a foreseeable harm; ACM 1.2 ('avoid harm') and BCS public interest clause both support liability. • Counter-argument (must be addressed for full marks): absence of misuse and prompt remediation are mitigating factors; courts may award nominal damages. Answers that only present one side lose 1 mark.	3
Q3 (CLO3) Reservation of function for IT professionals [8 marks]		
Q3a — Should life-critical IT development be reserved? [2 marks] (max 3 lines)		
	2 marks for a clear justified position within 3 lines: FOR (sample): Yes — Therac-25, Boeing 737 MAX, and London Ambulance disasters demonstrate that unregulated software development in safety-critical	2

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Q / Part	Marking Criteria & Model Answer Points	Marks
	domains causes preventable deaths. Reservation of function creates accountability and minimum competency standards. AGAINST (sample): No — Rapid technology change makes it impossible to define stable licensing criteria; self-regulation via industry standards (DO-178C, IEC 62304) and market accountability achieves similar safety goals more flexibly. <i>Deduct 1 mark if no justification is given. Deduct 1 mark if answer clearly exceeds 3 lines.</i>	
Q3b — Has this happened globally? [6 marks] (max 5 lines)		
	Award marks across these dimensions (up to 6 marks, within 5-line limit): • [2 marks] Partial yes: some jurisdictions require licensed engineers to sign off on embedded safety-critical systems (e.g., Canada's APEGBC, Australia's CPEng requirements for control systems). The USA PE licence is required for safety-critical infrastructure engineering. • [2 marks] Partial no: no globally unified reservation of function for software development exists; most countries rely on sector-specific standards and regulation (aviation DO-178C, medical IEC 62304, automotive ISO 26262) rather than licensing individual programmers. • [2 marks] Nuanced conclusion: the debate is ongoing; the software engineering licensing debate (Gotterbarn et al.) remains unresolved; current trend is toward domain-specific certification rather than blanket licensing. Answers demonstrating this nuance earn full marks.	6
Q4 (CLO3) Professional bodies: definition and roles [8 marks]		
Q4a — What is a professional body? [2 marks] (max 3 lines)		
	2 marks for a definition that contains all key elements within 3 lines: Model answer: A professional body is an organized association of practitioners in a specific discipline that sets and enforces standards of competence, ethics, and conduct for its members, and represents the interests of the profession to government and the public. Examples: BCS, ACM, IEEE-CS, PEC. Award 1 mark for a partial definition (membership + standards OR representation only). Award 2 marks for a full definition covering: organized membership + standard-setting + ethical oversight/representation. <i>Deduct 1 mark if answer clearly exceeds 3 lines.</i>	2
Q4b — Three roles/responsibilities of a professional body [6 marks] (max 3 lines each)		
	2 marks per role (1 mark: names role; 2 marks: names + explains with sufficient detail). Any three of the following qualify: 1. Setting Codes of Conduct [2 marks] Defines ethical standards and professional behaviour expectations; provides disciplinary mechanisms (investigation → sanction including expulsion) to enforce accountability among members. 2. Defining & Accrediting Education Standards [2 marks] Accredits degree programmes and sets training requirements to ensure graduates possess the knowledge base needed for professional practice (e.g., BCS-accredited degrees, NCEAC in Pakistan). 3. Disseminating Knowledge / Supporting CPD [2 marks] Organizes conferences, publishes peer-reviewed journals, and mandates Continuing Professional Development to keep members current with evolving technology and best practice.	6

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Q / Part	Marking Criteria & Model Answer Points	Marks
	4. Advising Government / Public Representation [2 marks] Provides expert policy recommendations to legislators (e.g., on cybercrime law, data protection), advocates for the profession, and builds public trust in IT practitioners. <i>Note: Deduct 1 mark per role if the explanation clearly exceeds 3 lines. Accept any well-explained role derived from lecture content.</i>	
TOTAL MARKS		40

CS4001 Professional Practices in IT — Marking Scheme